

Practical - 7

Practical Title: PHP Form Processing with Server-Side Validation and CSV/JSON File Storage

Problem Definition: Process submitted registration/contact form data using PHP. Validate and sanitize inputs on the server side and store records in CSV or JSON file format. Display success/error messages.

A. Theory

1. Is the form submitted using POST?

Answer: Yes, the form is submitted securely using method POST,

2. Are inputs validated and sanitized server side?

Answer: Yes, inputs are sanitized and strictly validated with regex and whitelists.

3. Is file writing handled safely?

Answer: Yes, file writing uses fputcsv(), atomic file_put_contents(), and bcrypt-hashed passwords.

4. Are success and error responses displayed clearly?

Answer: Yes, clear success/error messages are rendered via both styled UI pages and interactive JSON toasts.

B. Implementation/ Output/Result

Purpose	Processing
Registration handler	Regex(Name, Email, Phone), password complexity, bcrypt hashing
Contact message handler	Email filter, category, whitelist, length, bounds, Ticket ID gen
Admin dashboard	Parses stored files and renders responsive data tables
Client-side bridge	Asynchronous form dispatch, inline server errors, static fallback

C. Evidence of Work (Screenshots)

StudentHub Portal Registration

Registration Details

Full Name
Abc Def

Email Address
Abc.Def@gmail.com

Mobile Number
9120834756

Password
••••••••

Confirm Password
••••••••

Course
Computer Engineering

Description: Figure 7.1 details entering in registration page

Home Register Contact **Stored Records**

← Registration Form ← Contact Form

Student Registrations 4

Download CSV Download JSON

ID	Full Name	Email	Mobile	Course	Year	Gender	Password Hash	Registered At
STU-3ACB82	asdf	sgg@gmail.com	9087654321	Computer Engineering	2nd Year	Male	\$2y\$10\$IiHs611Fr...	2026-09-30 06:54:59
STU-A634D5	Abc Def	Abc.Def@gmail.com	9120834756	Computer Engineering	2nd Year	Male	\$2y\$10\$s2613MdgQh...	2026-09-30 19:23:22
STU-04F0A1	xyz abc	xyz.abc@gmail.com	9128374560	Information Technology	2nd Year	Male	\$2y\$10\$FNgyqqqfc...	2026-09-30 19:25:04
STU-D7A095	xyz abc	xyz.abc@gmail.com	9182736450	Computer Engineering	1st Year	Male	\$2y\$10\$n4ovAD7xPR...	2026-09-30 19:28:13

Contact Messages & Tickets 0

Description: Figure 7.2 Registration stored in both json and csv format.

D. GitHub Repository Link:

Link: <https://github.com/prep04-yl/StudentHub-Portal>

E. Challenges Faced:

Understanding how to run PHP scripts via an active web server instead of opening them directly in the browser, and coordinating regex validation rules with files permissions for CSV and JSON storage.

F. Learning Outcome:

I learned how to implement server-side form processing with safe file-based storage.

Faculty Remarks: (Keep this blank for faculty)